The Derivative Fallacy: Inverting Foundations

Andrew Martin

The derivative fallacy is the mistake of elevating a dependent abstraction above the primitive components that define it. When the derivative relation is treated as fixed by fiat, the primitives that once grounded it must be redefined elastically to preserve it. In this move, the relation ceases to approximate reality and collapses the very structure that originally gave it meaning. In any system of thought, derivatives cannot outrank their own definitions.

Examples

$$\textit{Velocity} = \frac{\textit{Distance}}{\textit{Time}}$$

In this example, velocity is defined as the amount of distance an object travels over some chosen interval of time. It is, by definition, derivative of the more primitive quantities of distance and time. If the velocity of an object is considered fixed by decree, any real changes in the system now must be absorbed by redefining either distance or time. The result is that the relation no longer describes motion. It conserves a ratio, which removes the necessary scaffolding for speaking to velocity at all.

If a horse at full gallop must always move at 40 miles per hour on a given track, we must conclude one of three things about the animal that crossed the finish line first: Its track was shorter, it experienced more seconds per minute, or it is not a horse.

Note: Even distance and time are mental constructs rather than ultimate primitives. They are selected as stable rulers that we use to measure change in the world. What counts as primitive is always relative to the framework. For example, time is usually treated as primitive, but in practice it is defined by oscillations of some other process. The derivative fallacy does not require absolute primitives. It only requires that within any given framework, a derivative cannot be elevated above a primitive that defines it.

$$\textit{Au price per ounce} = \frac{\textit{Dollars}}{\textit{Oz Gold}}$$

In this example, the price per ounce of gold is derived from the number of dollars required to purchase an ounce of gold. Dollars and ounces of gold are the primitives, and the entire purpose of this ratio is to track the relative value of one to the other. If the ratio is fixed by monetary policy at 35 dollars per ounce of gold, the ratio no longer describes real changes in purchasing power. When actual economic conditions inevitably vary, transactions must be restricted if arbitrage is to be prevented.

$$\textit{CPI} =$$

Current Basket of Goods Priced in Dollars
Baseline Basket of Goods Priced in Dollars

In this example, the consumer price index (CPI) aims to measure how the dollar cost of the same basket of goods changes over time. By definition, it is derivative of two primitives, the price of goods now and the price of the same goods then. Imagine the CPI is intentionally given a target; for instance, set to rise 2% per year. With the ratio fixed in advance, the only way to preserve the equals sign is to redefine the basket itself. Steaks become hamburger, butter becomes margarine.

Miles
Fuel Efficiency =
Gallons

In this example, fuel efficiency is derived by dividing distance traveled by fuel consumed. If the highway fuel economy of a vehicle is determined to be fixed at 50 miles per gallon, any real world deviation mathematically requires either miles or gallons to be redefined. The relation no longer measures fuel use, it preserves a ratio. In practice this would imply the odometer is wrong, not the assumption.Relation to Other Fallacies

The above examples show how mistakenly fixing the derivative forces the primitives that support it to become undefined, thereby dismantling the entire logical framework of that relationship. The derivative fallacy leans on and is adjacent to several well known fallacies. It may be seen as a special case of misplaced concreteness, or a specialized instance of category error, but is not completely reducible to either. It is defined by the inversion of dependency.

Aristotle’s Sophistical Refutations shows how arguments go wrong when assumptions are made or relations are misplaced. His catalog mostly covers linguistic traps like accident or equivocation. The derivative fallacy describes a specific kind of assumption about hierarchy, when a relation is used to override that which gave it meaning.

In the Categories, Aristotle distinguished substances (primitives) from relations (derivatives), making clear that the latter depend on the former. The derivative fallacy spells out the consequences of treating a relation as if it were a substance, and then bending the substance to make the relation fit.

Whitehead described the “fallacy of misplaced concreteness” as abstractions mistaken for solid realities. The derivative fallacy is a special case in which a derivative relation is supposed to be more concrete than its own primitives.

Korzybski echoed that same caution with “the map is not the territory.” The derivative fallacy presses further. Even among maps, order matters. Even abstract primitives cannot be usurped by derivatives that rely on them.

Ryle called it a “category mistake” to confuse something with a different kind of thing. The derivative fallacy shows that once miscategorized, the inputs themselves are bent to fit the output.Conclusion

This fallacy is not confined to philosophy or economics. It may already be shaping physics, where constants such as c are treated as primitives and the scaffolding of space and time is bent to preserve them. Other confusions in physics come from misplaced concreteness, but the derivative fallacy describes something specific, the inversion of order itself. If that pattern has taken root in science, the cost is structural, not cosmetic. Naming it clearly is a first step toward seeing where our abstractions have begun to dictate rather than describe.

References

Aristotle. (4th century BCE) Sophistical Refutations.
Aristotle. (4th century BCE) Categories.
Bridgman, P. W. (1927) The Logic of Modern Physics.
Duhem, P. M. M. (1954) The Aim and Structure of Physical Theory.
Korzybski, A. (1933) Science and Sanity: An Introduction to Non-Aristotelian Systems and General Semantics.
Lakatos, I. (1978) The Methodology of Scientific Research Programmes.
Ryle, G. (1949) The Concept of Mind.
Whitehead, A. N. (1925) Science and the Modern World.

Submitted to Zenodo on September 19, 2025
10.5281/zenodo.17163122
Contact: swolephoton@gmail.co